Unnath Chittimalla

Digital Design · DSP & Machine Learning · Biomedical Signal Processing · Product Engineering

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Education

International Institute of Information Technology Bangalore (IIIT-B) **Jul 2023 – Jul 2028**

Integrated M.Tech (Dual Degree B.Tech + M.Tech) in Electronics and Communication Engineering *CGPA: 3.49 / 4.00*

- **Coursework & Teaching:** VLSI Design, Computer Architecture & Processor Design, Digital Design, Embedded Systems; **Teaching Assistant** for **Signals & Systems** (Fall 2025, 150+ students).

Technical Skills

HDLs & Architecture: SystemVerilog, Verilog, VHDL, Vitis HLS, GPU ISA, Pipelining
EDA & FPGA Tools: Xilinx Vivado (OOC Synthesis, P&R, STA, VIO/ILA), Cadence Virtuoso (180nm GPDK), SkyWater 130nm PDK
Simulation, Power & Verification: Verilator, Icarus Verilog, cocotb, gem5, DRAMsim3, DRAMPower, LTspice, NGspice
Scripting, Programming & DSP: Python, Tcl (Vivado Batch Automation), C++, C, Bash/Linux, MATLAB, Git, PyTorch (Fixed-Point Quantization)

Research & IEEE Publications

Undergraduate Researcher, IIIT Bangalore **Jun 2025 – Present**

Advisor: Dr. Madhav Rao (Professor, ECE Department) *Bangalore, India*

- **First Author (IEEE EMBS BHI 2026):** *"Adaptive Lead-Field Residual Learning for Sparse-to-Dense EEG Reconstruction"* (*Code*); physics-informed sparse-to-dense EEG without structural MRI; RMSE 9.69 → **2.215** μ**V** ($R^2 = 0.955$), **77% sensor bandwidth/power cut**.
- **Co-Author (IEEE EMBS BHI 2025):** *"EEG-Based Surface EMG Reconstruction Using Deep Sequence Learning for Upper Limb Motor Activity"* (*Code*); EEG-only cortico-muscular decoding with CNN-LSTM; **2.11 ms latency**.

Selected Projects

TinyGPU: Minimal GPU in SystemVerilog (based on a public learning project) **Dec 2025 – Aug 2026**

SystemVerilog, Icarus Verilog, cocotb *GitHub | Video walkthrough*

- Studied and built a minimal multi-core GPU in **SystemVerilog**, following the public adam-maj/tiny-gpu learning project: dispatcher, per-core ALU/LSU pipelines, memory controller and a custom ISA; simulated matrix addition and multiplication kernels with **cocotb** (RTL design and simulation only).

OBC Distributed-Arithmetic DSP Core & HLS MLP **Oct 2025 – Jun 2026**

Verilog, VHDL, Vitis HLS, Artix-7 *GitHub*

- Multiplier-less Q15 inner-product core with **Distributed Arithmetic** and Offset Binary Coding (**50% fewer LUTs**) and a pipelined **Baugh-Wooley multiplier**, 2.1× faster critical path (1.473 → 0.704 ns).
- 16-bit fixed-point $784 \times 64 \times 10$ MLP in Vitis HLS & VHDL, validated on an Artix-7 (Basys 3) with Vivado VIO/ILA (*MLP repo*).

DDR4 PDN Decoupling & EDA Automation Flow **Apr 2026 – Jun 2026**

gem5, DRAMsim3, LTspice, Python *GitHub*

- **7-stage automation pipeline** (gem5, DRAMsim3, DRAMPower) producing SPICE waveforms, plus a batch-LTspice **Genetic Algorithm** optimizing an 86-capacitor network for an `LTM4632` regulator, suppressing anti-resonance spikes up to **1.6 GHz** within JEDEC ripple limits.

IMU-Only 2D Dead Reckoning with Sensor Fusion **Sep 2025**

Python, DSP, Kalman Filtering *GitHub*

- FFT-informed **Butterworth** and **Kalman filters** with a feature-driven Random Forest velocity regressor to mitigate accelerometer/gyroscope integration drift in 2D dead reckoning; **<2.5% distance error** against GPS ground truth.

GoalGuard (Co-Founder) **Jul 2025 – Present**

Kotlin, Android *Website · Play Store*

- Privacy-first distraction-blocking Android app with **20,000+ downloads across 20+ countries**; selected for **Y Combinator Startup School India**.

Honors & Achievements

- **Stage 2 Winner, NCIIPC & Startup India AI Grand Challenge (2026):** Coders@IIITB, PS-11 Hyperspectral Anomaly Detection (*Code*).
- **Top 10 Finalist, Meta Pragati AI for Impact (2025): 10,000+ teams**; low-latency multilingual RAG.
- **1st Place (5,000+ Participants), ICDEC VDVWC (2024):** Ranked #1 with a modified YOLOv10 for vehicle detection (*Code*).